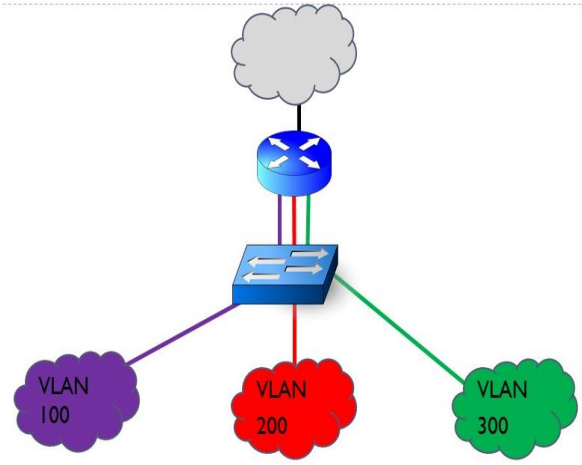


CLOUD COMPUTING COURSE

Lecture17-Network Virtualization in CC (6)

ABSTRACT

Lecture17Introduces Network in Cloud Computing

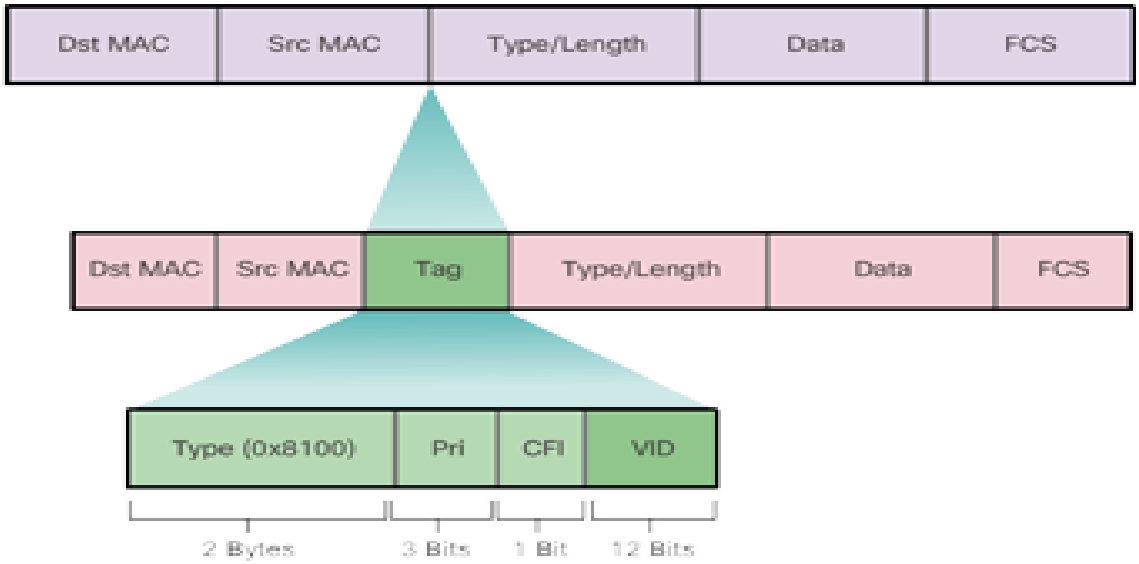
Instructor: Asma'a Khtoom

Tagging Ethernet Frames for VLAN Identification

When Ethernet frames are placed on a trunk, information about the VLANs to which they belong must be added.

This process, **called tagging,** is accomplished by using the **IEEE 802.1Q header**, specified in the IEEE 802.1Q standard. The 802.1Q header includes a **4-byte** tag inserted within the original Ethernet frame header, **specifying the VLAN to which the frame belongs**.

When the switch receives a frame on a port configured in access mode and assigned a VLAN, the switch **inserts a VLAN tag** in the frame header, recalculates the FCS, and sends the tagged frame out of a trunk port.



Benefits of VLANs

The primary benefits of using VLANs are as follows:

- **Security** - Groups that have sensitive data are separated from the rest of the network, decreasing the chances of confidential information breaches.

- **Cost reduction** - Cost savings result from reduced need for expensive network upgrades and more efficient use of existing bandwidth and links.

- **Better performance** - Dividing flat Layer 2 networks into multiple logical workgroups (broadcast domains) reduces unnecessary traffic on the network and boosts performance.

- **Shrink broadcast domains** - Dividing a network into VLANs reduces the number of devices in the broadcast domain.

Network Virtualization in CLOUD

Networks between VM's

Virtual Machines (VMs) run isolated from each other on one large host. However, as conventional computers communicate over a network, VMs can also be connected in different ways.

- A Virtual Machine (VM) is a combination of many virtual resources and among them is **one or more virtual** network interfaces.

- The hypervisor or Virtual Machine Manager (VMM), which is the software layer responsible for managing VMs, prepares these network interfaces and make them appear to the guest operating system, which runs inside the VM, as conventional Network Interface Cards (NICs) with MAC addresses assigned on creation.

Virtual Network Devices in Cloud

Constructing a network requires devices other than the NICs such as switches, routers, Dynamic Host Configuration Protocol (DHCP) server, etc.

In the virtualized environment, the VMM has virtual network devices that simulate the physical ones. Some of these virtual network devices are:

- Virtual switch
- Bridge
- Virtual Host Adapter
- Network Address Translation (NAT) device
- DHCP server

VM Network Modes

The VMM uses previous virtual network devices in different combinations to allow scenarios where VMs connect with each other and with the external network; the host machine has access to. The most popular three networking plots of VMs are:

1. NAT networking
2. Host-only networking
3. Bridged networking

NAT - Network Address Translation

In this mode, the VMM creates a connection between the virtual network interface of the VM and a NAT device which is in turn connected to a DHCP server as demonstrated in Figure-1.

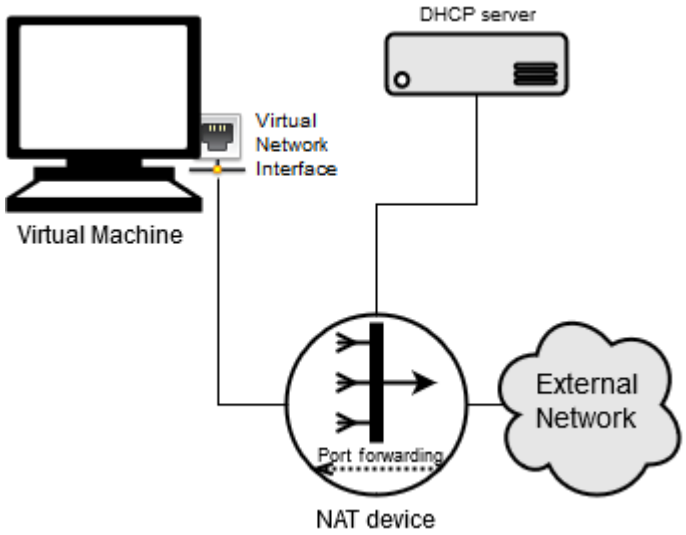


Figure-1: Network Address Translation (NAT) Networking

The VM obtains a private IP address from the DHCP server upon starting up, and the VMM maps this address to a TCP port number on the host machine. Traffic originated from the VM will be seen on the external network as if it was coming from the host machine's IP address, not the VM's. The NAT mode permits access to a VM from the external network by implementing the **port forwarding** feature.

Host-only Networking

This mode is useful when the user wishes to connect a set of VMs, in addition to the host platform, to an isolated network. In this scenario, the VMM creates a virtual host adapter that acts as a NIC within the host machine and connects it, in addition to the other VMs, to a virtual switch as shown in Figure-2.

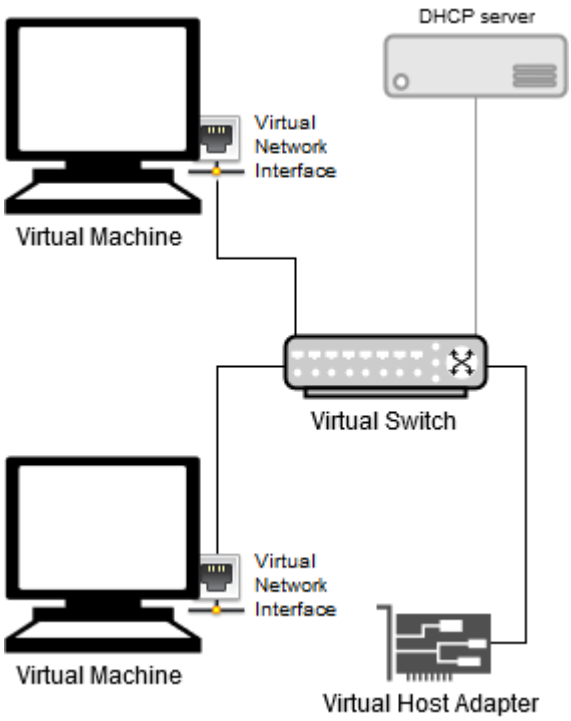


Figure-2: Host-only Networking

The user may let the machines in the host-only network to obtain their configurations from an optional DHCP server that can be attached to the network or configure them manually. The VMs in this scenario neither appear on the external network, nor can they access it.

Bridged Networking

This scenario provides a solution that allows VMs to be exposed to the external network the host machine connects to, as illustrated in Figure-3. The VM will appear as an individual platform on the external network and get its configuration in the same way the host machine does.

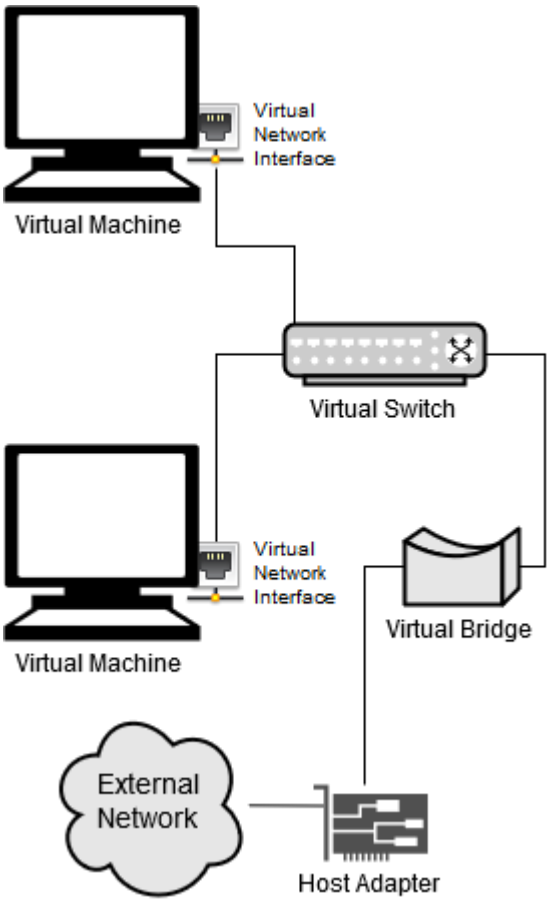


Figure-3: Bridged Networking

The Table below show the connections provided by different network modes in the cloud environment for the cloud VM's.

Table-1: An overview of networking modes

	VM↔Host	**VM↔VM**	**VM→External network**	**External network→VM**	**VM's IP Visiblity to the External Network**
NAT	No	No	Yes	Yes(Port forwarding)	Hidden
Host-only	Yes	Yes	No	No	Hidden
Bridged	Yes	Yes	Yes	Yes	Visible